MIL-G-17787C(YD)
10 February 1983
SUPERSEDING
MIL-G-17787B(YD)
7 May 1974

MILITARY SPECIFICATION

GAGES, DIAL INDICATING, FURNACE DRAFT

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers single/multiple-pointer draft gages suitable for the measurement and indication of draft and low pressures in boiler furnaces.

1.2 Classification. The gages shall be of the following type and styles, as specified (see 6.2):

Type I - Diaphragm or bellows.
Style 1 - Straight-line scale.
Style 2 - Circular scale.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116 - Preservation, Methods of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.
MIL-STD-794 - Parts and Equipment, Procedures for Packaging and Packing of.

(Copies of specifications and standards, required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The equipment shall consist of one or more gages arranged for mounting. Each gage shall consist of an actuating system and an indicating system. The individual gages shall be equipped with all parts, components, and accessories as specified herein and as required for satisfactory operation. Type I instruments shall be mechanically actuated, without liquid means, by the use of bellows or diaphragms.

3.2 First article. When specified (see 6.2), the contractor shall furnish a gage for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Standard commercial product. The dial indicating gages shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the gages being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

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3.5 Construction.

3.5.1 Type I. The pressure responsive element shall be of proven diaphragm quality material and housing construction suitable for the service. The material shall be of such a nature, or so treated, that it will not harden, crack, or otherwise deteriorate under normal operating conditions. The elements shall be so designed that progressive set and consequent drift from calibration under normal operating conditions shall be imperceptible.

3.5.2 Style 1. Style 1 instruments shall consist of one or more gages as specified (see 6.2). The straight-line scale shall be either flat or curved into a segment of a cylinder. Flat scales for multiple gages shall be positioned side by side in the same plane. Curved scales for multiple gages shall be mounted parallel with and alongside each other so that they form an arc of a cylindrical surface not greater than 90°. The gages comprising the instrument shall be housed in a single case or each gage shall be housed in a separate case with the separate gages arranged side by side and enclosed by a single bezel or frame.

3.5.3 Style 2. Style 2 instruments shall consist of one or more gages, as specified (see 6.2), housed within a single case. The pointer(s) of the dial scales shall move in a circular arc that shall not exceed 180° in a plane normal to the line of vision of the observer.

3.5.4 Indicating system. Each gage shall be provided with a scale, a pointer, and a magnifying linkage or gearing between the pressure-sensitive element and the pointer.

3.5.4.1 Scale. Unless otherwise specified (see 6.2), the units of pressure used for scale graduation shall be inches of water. Subdivision graduations for all style 1 and style 2 scales, 8 inches and longer, shall be no larger than those listed in table I for the maximum scale ranges listed. Graduation markings and scale numerals shall be of appropriate proportion, size, and location so that indicated values can be readily determined. The numerals and graduations next larger than those listed in table I shall be suitable for reading by a person with normal vision from a distance of 20 feet.

TABLE I. Scale subdivisions.

Maximum scale range (± inches of water)	Subdivision (inches of water)
0.5	0.01
1.0	0.02
2.0	0.05
5.0	0.10
15.0	0.50
40.0	1.00

Scale backgrounds shall be white and markings shall be of dark color in contrast with the background. Style 1 scales shall be not more than 2 inches wide and not less than 8 inches long. When specified (see 6.2), each scale on style 1 instruments shall be marked to identify its service. Unless otherwise specified (see 6.2), style 2 scales shall be not less than 10, 9, and 8 inches long for gages with 1, 2, and 3 pointers, respectively. When style 2 scales shorter than 8 inches are specified, the shortest scale length shall be not

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less than 5 inches. Graduation markings for scales shorter than 8 inches shall be not greater than twice the subdivision requirements listed in table I for the applicable scale range.

3.5.4.2 Pointer. The pointer shall be of a contrasting color with the scale background and shall have a fine, narrow point or arrow head to permit accurate readings. A means shall be provided for bringing each pointer to zero without the need for special purpose tools. Suitable stops shall be provided to prevent overtravel of the pointer.

3.5.4.3 Mechanical linkage. The linkage shall magnify the movement of the pressure-sensitive element to the greatest extent practicable for the accuracy required over the range of the instrument. The mechanism shall have suitable means provided for required adjustments to calibrate the instrument. Bearings shall be of the knife edge, antifriction, ring-type jewel, or bushing of hard noncorrosive type. Antifriction bearings shall be of the lubricated-for-life type.

3.5.5 Illumination. Unless otherwise specified (see 6.2), the scales shall be illuminated. The light shall be suitable for operation of the power supply, as specified (see 6.2), shall illuminate the scales clearly without glare, and shall be so arranged that the heating effect will cause no variation in the calibration of the instrument.

3.5.6 Alarm. When specified (see 6.2), an audible alarm, electrically actuated, shall be provided to warn of maximum or minimum pressure conditions or both. The alarm shall be sounded by an electrical switch actuated by the mechanical linkage, directly by the draft or low pressure. The switch shall be of a type which will assure tripping of the alarm circuit at the conditions specified. The alarm circuit shall be suitable for operation on the power supply as specified (see 6.2).

3.5.7 Case. The scale, pointer, linkage, and actuating mechanism shall be completely enclosed in a case with a glass or clear plastic front. The glass or clear plastic front shall permit unobstructed observation of the pointer over the complete scale range. The case shall be sheet metal, aluminum, or other suitable material adequately protected against corrosion and shall be dusttight. The arrangement of the case shall be such that all piping, tubing, or electrical connections are accessible without removal of the case from its mount. Cases shall be suitable for surface, flush, or semiflush mounting as specified (see 6.2).

3.5.8 Pressure connections. Gages designed to measure pressures above atmospheric pressure shall be provided with a single pressure connection to the diaphragm, or bellows. The pressure connection shall be provided with a 3-way stopcock which will permit checking of the zero setting and calibration at any time without breaking service connections. Gages designed to measure differential pressure shall be provided with two pressure connections, one to each side of the pressure sensitive element. Stopcocks shall be provided on each of these connections to permit checking the calibration and equalizing the pressure between both sides of the pressure sensitive element without breaking service connections. The stopcocks for the inch differential connections may be either an integral or fabricated manifold arrangement. A flared union coupling, 0.250-inch National taper pipe (NPT) female connection, or a nylon lined compression nut shall be provided on all service connections and shall be 0.250-inch tubing size, unless otherwise specified (see 6.2).

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3.6 Performance.

3.6.1 Range. Each indicating unit shall be suitable for the measurement of pressures over the specified range. The specified range shall be completely contained within the calibrated range of the instrument and shall encompass no more than 80 percent of it. Gages shall operate within the following range, as specified (see 6.2):

Type I ±40 inches of water, based on atmospheric pressure as the reference point.
±12 inches of water, based on atmospheric pressure as the reference point.
From 0 to 24 inches of water above atmospheric pressure.

3.6.2 Accuracy. The gages shall be designed and constructed to insure accuracy within ± 2 percent of the full scale reading in ambient temperatures varying from 35° Fahrenheit (F) to 135°F. The operating mechanism shall withstand, without affecting gage accuracy or damage to the instrument, surges equal to 100 percent above or below the entire scale range (see 4.5.2).

3.6.3 Calibration. The gages shall be calibrated prior to delivery and shall require no further calibration or adjustment other than setting the pointer to zero.

3.7 Threads. Screw threads for all securing or attaching devices and threaded parts shall be in conformance with FED-STD-H28 as follows:

a. Machine threads shall be the standard national coarse or fine thread series.
b. Pipe threads shall be the standard NPT thread series.

3.8 Interchangeability. All gages of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.9 Treatment and painting. Unless otherwise specified (see 6.2), the gage shall be treated and painted in accordance with the manufacturer's standard practice. All surfaces of the gage other than corrosion-resisting steel shall be protected against corrosion and shall present a neat appearance.

3.10 Identification marking. Identification shall be permanently and legibly marked directly on each gage or on a corrosion-resisting metal plate securely attached to the gage by the manufacturer. Identification shall include the manufacturer's model and serial number, name and trademark to be readily identifiable to the manufacturer.

3.11 Workmanship. The quality of workmanship shall be such as to produce gages that are in accordance with the requirements of this specification and constructed to insure proper functioning of all parts of the units.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified

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in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one gage when a first article is required (see 3.2 and 6.4). This inspection shall include the examination of 4.4 and the tests of 4.5. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. All gages offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.3.1 Sampling for examination. Examination shall be based on inspection level II and an acceptable quality level (AQL) of 1.5 percent defective.

4.3.2 Sampling for tests. Tests shall be based on inspection level II and an AQL of 4.0 percent defective.

4.4 Examination. Each sample selected shall be examined for defects listed in table II.

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TABLE II. Classification of defects.

Classification	Defects	Requirement paragraph
Critical	None	
Major:		
101	Style 1 curved scales form an arc greater than 90°	3.5.2
102	Style 2 pointers move in a circular arc greater than 180°	3.5.3
103	Scale not legible	3.5.4.1
104	Scale not in accordance with table I	
105	No means provided to calibrate the gage	3.5.4.3
106	Alarm will not actuate	3.5.6
Minor:		
201	Scale background color not as specified	3.5.4.1
202	Pointer not a contrasting color with scale background	3.5.4.2
203	Inadequate illumination	3.5.5
204	Plastic front discolored	3.5.7
205	Connections not as specified	3.5.8
206	Component parts and accessories not interchangeable	3.8
207	Treatment and painting not as specified	3.9
208	Identification marking not as specified	3.10

4.5 Tests. Each gage selected in accordance with 4.3.2 shall be tested and any gage failing to pass the following tests, as applicable, shall be rejected.

4.5.1 Leak test. The gage shall be subjected to 4 half-hour periods of pressure or draft (vacuum) alternated between the lowest and highest points of the specified range. The gage shall show no deviation in excess of 2 percent of the actual pressure during or after the test (see 3.6.1).

4.5.2 Accuracy test. The gage shall be subjected to accuracy tests at the cardinal points over the specified range at temperatures varying from 35°F to 135°F with a standard absolute pressure mercury gage or precision manometer. The gage shall show no deviation greater than 2 percent from the correct pressure or draft at the points tested (see 3.6.2).

4.5.3 Surge test. The gage shall be subjected to surges equal to 100 percent above or below the entire scale range. The gage shall be rejected if the mechanism becomes damaged or its accuracy is effected during this test (see 3.6.2).

4.6 Packaging inspection. The preservation, packing, palletization, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

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5.1.1 Level A. Each gage shall be packaged method III as specified in MIL-P-116 in a fiberboard box in accordance with PPP-B-636, class weather-resistant.

5.1.2 Commercial. The gage shall be preserved in accordance with the contractor's standard practice in a manner to prevent deterioration and damage.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Levels A and B. Packing shall be accordance with MIL-STD-794. Containers shall be selected from table I for the appropriate level.

5.2.2 Commercial. The gage shall be prepared for shipment in a manner which will insure arrival at destination in a satisfactory condition. Preparation for delivery shall comply with applicable carrier rules and regulations.

5.3 Palletization. Material shall be palletized in accordance with MIL-STD-147 when the following criteria is met:

a. Load to consist of four or more unskidded containers; and,
b. Load shall utilize a minimum of 80 percent of the pallet base.

5.4 Marking. Marking shall be accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Gages covered by this specification are intended for measuring and indicating draft and low pressures in furnaces at onshore installations.

6.1.1 Type. Type I instruments can be supplied for either direct pressure or differential pressure measurement.

6.1.2 Styles. Style 1 instruments are suitable for any number of gages required. Style 2 instruments are offered with up to three gaging units. Style 2 instruments with scale lengths of 10, 9, and 8 inches (see 3.5.4.1) are suitable for permanent installation in large power plants. Smaller style 2 scales are available down to 5 inches in length for installation on small package boilers and where instrument panel space is a prime factor.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specfication.
b. Type and style required (see 1.2).
c. Issue of document required, if other than as specified (see 2.1.1).
d. When first article is required for inspection and approval (see 3.2 and 6.4).
e. Number of gage units housed in a single case required (see 3.5.2 and 3.5.3).
f. Units of pressure for scale graduations, if other than specified (see 3.5.4.1).
g. Scale identification for each of style 1 scales, if required (see 3.5.4.1).
h. Length of style 2 scales, if shorter than those specified (see 3.5.4.1).
i. When scale illumination is not required (see 3.5.5).

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j. Power supply (voltage, phase, and frequency) for illumination and alarm circuit as applicable (see 3.5.5 and 3.5.6).
k. Audible electric alarm, if required, and the pressure conditions for operation (see 3.5.6).
l. Type of mounting arrangement for case (see 3.5.7).
m. Size of union coupling, if other than specified (see 3.5.8).
n. Pressure range for type I, indicating unit (see 3.6.1).
o. Treatment and painting, if other than specified (see 3.9).
p. Level of preservation and level of packing required (see 5.1 and 5.2).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a Contract Data Requirements List (DD Form 1423), and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved DD Form 1423 incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one gage. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Cross-reference. One change has been made in requirements under the specification. Requirement for Type II gage has been eliminated and all requirements specified under Type I.

MIL-G-17787B(YD)	MIL-G-17787C(YD)
Type I - Diaphragm or Bellows	Type I - Diaphragm or Bellows
Type II - Submerged Bell	Eliminated

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Review activity:

DLA - GS

Preparing activity:

Navy - YD

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